Ch.2: Probability & HMM

Probability Basics

Axioms: $P(A) \geq 0$; $P(\Omega) = 1$; disjoint A, B: $P(A \cup B) = P(A) + P(B)$.

Bayes Rule:

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Total Probability: $P(B) = \sum_i P(B|A_i)P(A_i)$

Conditional Independence: $X \perp Y|Z \Rightarrow P(X,Y|Z) = P(X|Z)P(Y|Z)$

Random Variables

Expectation: $\mathbb{E}[X] = \sum_x x\,p(x)$

Variance: $\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2]$

Covariance: $\mathrm{Cov}(X,Y) = \mathbb{E}[(X-\mu_X)(Y-\mu_Y)]$

Markov Chain

State: x_k. **Markov property**: $p(x_k|x_{0:k-1}) = p(x_k|x_{k-1})$

Transition: $p(x_k|x_{k-1}) = T_{x_{k-1}\to x_k}$

Stationary dist.: $\pi = T^T\pi$

Hidden Markov Model (HMM)

- States x_k (hidden), observations y_k (visible)
- **Prior**: $p(x_0)$; **Transition**: $p(x_k|x_{k-1})$; **Emission**: $p(y_k|x_k)$
- Joint: $p(x_{0:T}, y_{0:T}) = p(x_0)\prod_k p(y_k|x_k)p(x_k|x_{k-1})$

Bayes Filter (Forward)

Predict: $p(x_k|y_{0:k-1}) = \sum_{x_{k-1}} p(x_k|x_{k-1})\,p(x_{k-1}|y_{0:k-1})$

Update: $p(x_k|y_{0:k}) \propto p(y_k|x_k)\,p(x_k|y_{0:k-1})$

Define $\alpha_k(x_k) = p(x_k, y_{0:k})$:

$$\alpha_k(x_k) = p(y_k|x_k)\sum_{x_{k-1}} p(x_k|x_{k-1})\alpha_{k-1}(x_{k-1})$$

$p(y_{0:T}) = \sum_{x_T}\alpha_T(x_T)$ (Forward algorithm)

Backward Algorithm

$\beta_k(x_k) = p(y_{k+1:T}|x_k)$; $\beta_T(x_T) = 1$

$$\beta_{k-1}(x_{k-1}) = \sum_{x_k} p(y_k|x_k)p(x_k|x_{k-1})\beta_k(x_k)$$

Smoothing: $p(x_k|y_{0:T}) \propto \alpha_k(x_k)\beta_k(x_k)$

Viterbi (MAP Sequence)

Find $\hat{x}_{0:T} = \arg\max p(x_{0:T}|y_{0:T})$

$\delta_k(x_k) = \max_{x_{0:k-1}} p(x_{0:k}, y_{0:k})$

$$\delta_k(x_k) = p(y_k|x_k)\max_{x_{k-1}} p(x_k|x_{k-1})\delta_{k-1}(x_{k-1})$$

Traceback: $\psi_k(x_k) = \arg\max_{x_{k-1}}[\cdot]$

Ch.3: Kalman & Particle Filters

Gaussian

$\mathcal{N}(x;\mu,\Sigma) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}}\exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$

Linear transform: $y = Ax + b$, $x \sim \mathcal{N}(\mu,\Sigma) \Rightarrow y \sim \mathcal{N}(A\mu + b,\ A\Sigma A^T)$

Marginalization: $\begin{bmatrix} x \\ y \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix}\mu_x \\ \mu_y\end{bmatrix}, \begin{bmatrix}\Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy}\end{bmatrix}\right) \Rightarrow x|y \sim \mathcal{N}(\mu_x + \Sigma_{xy}\Sigma_{yy}^{-1}(y-\mu_y),\ \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx})$

Kalman Filter (KF)

Model: $x_{k+1} = F_k x_k + B_k u_k + w_k$, $w_k \sim \mathcal{N}(0, Q_k)$

$y_k = H_k x_k + v_k$, $v_k \sim \mathcal{N}(0, R_k)$

Predict:

$$\hat{x}_{k|k-1} = F_{k-1}\hat{x}_{k-1|k-1} + B_{k-1}u_{k-1}$$
$$P_{k|k-1} = F_{k-1}P_{k-1|k-1}F_{k-1}^T + Q_{k-1}$$

Update (Kalman gain K_k):

$$K_k = P_{k|k-1}H_k^T(H_kP_{k|k-1}H_k^T + R_k)^{-1}$$
$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(y_k - H_k\hat{x}_{k|k-1})$$
$$P_{k|k} = (I - K_kH_k)P_{k|k-1}$$

Extended KF (EKF)

Nonlinear model: $x_{k+1} = f(x_k, u_k) + w_k$, $y_k = h(x_k) + v_k$

Linearize at current estimate:

$$F_k = \left.\frac{\partial f}{\partial x}\right|_{\hat{x}_{k|k}}, \quad H_k = \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_{k|k-1}}$$

Use KF equations with F_k, H_k (Jacobians), but propagate mean through nonlinear f, h.

Unscented KF (UKF)

Choose **sigma points** $\{\mathcal{X}_i\}$ to capture mean & covariance:

$$\mathcal{X}_0 = \mu, \quad \mathcal{X}_i = \mu + (\sqrt{(n+\lambda)\Sigma})_i, \quad \mathcal{X}_{n+i} = \mu - (\sqrt{(n+\lambda)\Sigma})_i$$

$\lambda = \alpha^2(n+\kappa) - n$. Weights: $W_0^m = \frac{\lambda}{n+\lambda}$, $W_i^m = W_i^c = \frac{1}{2(n+\lambda)}$.

Propagate through nonlinear f; recover predicted mean & cov from weighted sigma points. No Jacobians needed.

Particle Filter

Represent $p(x_k|y_{0:k})$ by particles $\{x_k^{(i)}, w_k^{(i)}\}$.

Algorithm:

1. **Predict**: Sample $x_k^{(i)} \sim p(x_k|x_{k-1}^{(i)})$
2. **Weight**: $w_k^{(i)} \propto p(y_k|x_k^{(i)})$; normalize
3. **Resample**: Draw N particles with replacement per weights

Importance Sampling: use proposal $q(x)$; weight $\propto p(x)/q(x)$.

Degeneracy: use **ESS**$= 1/\sum(w^{(i)})^2$; resample when ESS$< N/2$.

Ch.4: Geometry & Mapping

SO(3) — Rotation Matrices

$R \in SO(3)$: $R^TR = I$, $\det R = 1$, 3 dof.

Composition: $R_{AC} = R_{AB}R_{BC}$; Inverse: $R^{-1} = R^T$.

Skew-symmetric: $[\omega]_\times = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$, $[\omega]_\times v = \omega \times v$.

Rodrigues' Formula

Rotate by angle θ about unit axis $\hat{k}$:

$$\boxed{R = I + \sin\theta\,[\hat{k}]_\times + (1-\cos\theta)\,[\hat{k}]_\times^2}$$

Equivalently: $R = e^{\theta[\hat{k}]_\times}$ (matrix exponential).

Key: $[\hat{k}]_\times^3 = -[\hat{k}]_\times$, so Taylor series closes at 3 terms.

Log map: $\theta = \cos^{-1}\left(\frac{\mathrm{tr}(R)-1}{2}\right)$; $[\hat{k}]_\times = \frac{R-R^T}{2\sin\theta}$.

Euler Angles & Gimbal Lock

$R = R_z(\psi)R_y(\phi)R_x(\theta)$ (ZYX order).

Gimbal lock: when $\phi = \pm 90$, lose one dof (singularity).

Lie Group / Algebra

$\mathfrak{so}(3)$ = skew-symmetric matrices $\approx \mathbb{R}^3$.

$\exp : \mathfrak{so}(3) \to SO(3)$, $\log : SO(3) \to \mathfrak{so}(3)$.

$\mathrm{Ad}_R\,\omega = R\omega$ (adjoint action).

SE(3) — Rigid Body

$T = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} \in SE(3)$; $T^{-1} = \begin{bmatrix} R^T & -R^Tt \\ 0 & 1 \end{bmatrix}$.

Transform point: $p' = Rp + t$.

Quaternions

$q = (w, \mathbf{v}) = w + ix + jy + kz$, $|q| = 1$ for unit quat.

Rotation: $q = (\cos\frac{\theta}{2}, \sin\frac{\theta}{2}\hat{k})$.

Apply: $p' = q \otimes (0,p) \otimes q^{-1}$; $q^{-1} = (w, -\mathbf{v})$.

Compose: $q_1 \otimes q_2$ (Hamilton product); note $q \equiv -q$.

Occupancy Grid

Cell probability $p_k = p(\mathrm{occ}|y_{0:k})$.

Log-odds: $l_k = \log\frac{p_k}{1-p_k}$; update: $l_k = l_{k-1} + \log\frac{p(y_k|\mathrm{occ})}{p(y_k|\mathrm{free})} - l_0$

Sonar/LiDAR model: cells before hit $\to$ free; hit cell $\to$ occupied.

Recover prob: $p = \frac{e^l}{1+e^l} = \sigma(l)$.

SDF, NeRF

SDF: $d(x)$ = signed distance to nearest surface; $d = 0$ on surface.

NeRF: volume radiance field; render via ray marching:

$$C = \int_0^\infty T(t)\,\sigma(r(t))\,c(r(t), \mathbf{d})\,dt$$

$T(t) = \exp(-\int_0^t \sigma(s)ds)$ (transmittance).

Discrete:

$$\hat{C} = \sum_i T_i(1-e^{-\sigma_i\delta_i})c_i, \quad T_i = \prod_{j<i} e^{-\sigma_j\delta_j}$$

Ch.5: DP & Dijkstra

Optimal Control Problem

Dynamics: $x_{k+1} = f_k(x_k, u_k)$, $x_k \in \mathcal{X}$, $u_k \in U$.

Total cost:

$$J(x_0; u_{0:T-1}) = q_f(x_T) + \sum_{k=0}^{T-1} q_k(x_k, u_k)$$

Goal: $u^*_{0:T-1} = \arg\min_u J(x_0; u_{0:T-1})$

Dijkstra's Algorithm

Find shortest path from source s in weighted graph $G = (V, E)$, edge costs ≥ 0.

Backward Dijkstra (from goal g):

1. Init: $\text{dist}(g) = 0$; $\text{dist}(v) = \infty \; \forall v \neq g$; $S = \emptyset$.
2. Repeat: pick $v \notin S$ with min dist; add to S.
3. For each neighbor u of v: if $\text{dist}(u) > \text{dist}(v) + c(v, u)$:

   $\text{dist}(u) \leftarrow \text{dist}(v) + c(v, u)$; $\quad \text{prev}(u) \leftarrow v$
4. Until $S = V$ or goal reached.

Complexity: $\mathcal{O}((m + n) \log n)$ with binary heap (m=edges, n=nodes).

Key: non-negative edge weights required (Dijkstra fails with negative weights $\rightarrow$ use Bellman-Ford).

Connection to DP: On finite graph, Dijkstra solves the DP: $J^*(v) = \min_{u:(v,u)\in E}\{c(v, u) + J^*(u)\}$ with $J^*(g) = 0$.

Principle of Dynamic Programming

Bellman optimality: $u^*_{k:T-1}$ optimal $\Rightarrow u^*_{k+1:T-1}$ optimal from x_{k+1}.

DP recursion (backward in time):

$$\boxed{J^*_k(x) = \min_{u_k \in U}\{q_k(x, u_k) + J^*_{k+1}(f_k(x, u_k))\}}$$

$J^*_T(x) = q_f(x)$. Solve backward $k = T - 1, \ldots, 0$.

Q-factor (action-value):

$$Q^*_k(x, u) = q_k(x, u) + J^*_{k+1}(f_k(x, u))$$

$J^*_k(x) = \min_u Q^*_k(x, u)$; optimal policy $\mu^*_k(x) = \arg\min_u Q^*_k(x, u)$.

Note: Viterbi $\equiv$ DP for MAP sequence in HMM.

Stochastic DP

Model: $x_{k+1} = f_k(x_k, u_k, \epsilon_k)$, ϵ_k noise.

Stochastic DP recursion:

$$J^*_k(x) = \min_{u_k \in U} \mathbb{E}_\epsilon\left[q_k(x, u_k) + J^*_{k+1}(f_k(x, u_k, \epsilon))\right]$$

Infinite-Horizon DP & Value Iteration

Discounted cost: $J(x_0; \pi) = \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k q(x_k, \mu(x_k))\right]$, $\gamma \in (0, 1)$.

Bellman equation:

$$\boxed{J^*(x) = \min_{u \in U} \mathbb{E}_\epsilon[q(x, u) + \gamma J^*(f(x, u, \epsilon))]}$$

Value Iteration:

$$J^{(i+1)}(x) = \min_{u \in U} \mathbb{E}_\epsilon\left[q(x, u) + \gamma J^{(i)}(f(x, u, \epsilon))\right]$$

Converges: $\|J^{(i)} - J^*\|_\infty \leq \frac{\gamma^i}{1-\gamma}\|J^{(0)} - J^*\|_\infty$.

Q-iteration:

$$Q^{(i+1)}(x, u) = \mathbb{E}_\epsilon\left[q(x, u) + \gamma \min_{u'} Q^{(i)}(f(x, u, \epsilon), u')\right]$$

Policy Evaluation & Iteration

Fixed policy π (matrix form):

$$J^\pi(x) = q^\pi(x) + \gamma \sum_{x'} T^\pi(x, x') J^\pi(x')$$

$$\Rightarrow J^\pi = (I - \gamma T^\pi)^{-1} q^\pi$$

$(I - \gamma T^\pi)$ invertible since $\gamma < 1$ and T^π is stochastic.

Policy Iteration:

1. **Evaluate**: solve $J^\pi = (I - \gamma T^\pi)^{-1} q^\pi$ (or iterate)
2. **Improve**: $\pi'(x) = \arg\min_u \mathbb{E}[q(x, u) + \gamma J^\pi(f(x, u, \epsilon))]$
3. Repeat until $\pi' = \pi$ (convergence guaranteed, finite steps).

Note: Policy Iteration $\leq |U|^{|X|}$ steps; often faster than Value Iteration.

Finite vs Infinite Horizon Summary

	Finite Horizon	Infinite Horizon
Policy	Time-varying $\mu_k(x)$	Stationary $\mu(x)$
Solve	Backward DP	Value/Policy Iter.
Discount	Not needed	$\gamma \in (0, 1)$ needed
Bellman	Recursion	Fixed point eq.

Key Relationships

- Dijkstra = DP on graph (backward from goal)
- Viterbi = DP for MAP in HMM (Ch.2)
- KF = optimal for linear-Gaussian DP
- Q-factor $\leftrightarrow$ Q-function in RL

Quick Reference Formulas

Gaussian product: $\mathcal{N}(\mu_1, \Sigma_1) \cdot \mathcal{N}(\mu_2, \Sigma_2) \propto \mathcal{N}(\mu^*, \Sigma^*)$
$\Sigma^{*-1} = \Sigma_1^{-1} + \Sigma_2^{-1}$; $\mu^* = \Sigma^*(\Sigma_1^{-1}\mu_1 + \Sigma_2^{-1}\mu_2)$

Matrix inversion lemma: $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$

Geometric series: $\sum_{k=0}^{\infty} \gamma^k = \frac{1}{1-\gamma}$ for $|\gamma| < 1$

Log-sum-exp trick: $\log \sum_i e^{a_i} = a^* + \log \sum_i e^{a_i - a^*}$

SO(3) properties:
$\exp([\omega]_\times)$ = rotation by $\|\omega\|$ about $\hat{\omega} = \omega/\|\omega\|$
$\text{tr}(R) = 1 + 2\cos\theta$; $R^{-1} = R^T$; $\det R = 1$

Quaternion norm: $|q_1 \otimes q_2| = |q_1||q_2|$; $q^{-1} = q^*/|q|^2$

Bellman contraction: $\|BJ_1 - BJ_2\|_\infty \leq \gamma\|J_1 - J_2\|_\infty$ (B = Bellman operator)